Sales Data Analysis - Retail Store

Simple Data Analysis Project using Python (Pandas & Matplotlib)

1. Introduction

Ek retail store ka 6 mahine (Jan-Jun) ka sales data banaya gaya hai jisme 3 product categories hain: Electronics, Clothing aur Grocery. Is data par simple analysis karke business ke liye useful insights nikale gaye hain.

2. Dataset

Dataset me 18 rows aur 5 columns hain: Month, Category, Units_Sold, Revenue (Rs) aur Profit (Rs).

Month	Category	Units_Sold	Revenue	Profit
Jan	Electronics	40	200000	30000
Feb	Electronics	35	175000	24500
Mar	Electronics	50	250000	40000
Apr	Electronics	45	225000	31500
May	Electronics	55	275000	44000
Jun	Electronics	60	300000	51000
Jan	Clothing	120	96000	19200
Feb	Clothing	110	88000	15840
Mar	Clothing	130	104000	22880
Apr	Clothing	140	112000	25760
May	Clothing	125	100000	20000
Jun	Clothing	150	120000	28800
Jan	Grocery	400	80000	8000
Feb	Grocery	420	84000	9240
Mar	Grocery	410	82000	7380
Apr	Grocery	430	86000	9460
May	Grocery	450	90000	9900
Jun	Grocery	460	92000	11040

3. Question

Upar diye gaye sales data ke basis par neeche ke questions ka answer nikaliye:

Q1. Har category ka total Revenue aur total Profit kitna hai?

Q2. Kis month me sabse zyada Revenue aaya? (Monthly trend bhi dekhiye)

Q3. Kis category ka Profit Margin % sabse zyada hai?

4. Solution (Step by Step)

Step 1: Pandas aur Matplotlib libraries import karo.

Step 2: Data ko dictionary me likh kar DataFrame banao.

Step 3 (Q1): `groupby('Category')` se Revenue aur Profit ka sum nikalo.

Step 4 (Q2): `groupby('Month')` se monthly Revenue nikalo aur `idxmax()` se best month dhundo.

Step 5 (Q3): Profit Margin % = (Profit / Revenue) x 100 formula lagao.

Step 6: Bar chart aur line chart se results ko visualize karo.

5. Python Code

```
# Simple Sales Data Analysis - Retail Store
import pandas as pd
import matplotlib.pyplot as plt

# ---------- Step 1: Data banana ----------
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"] * 3,
    "Category": ["Electronics"] * 6 + ["Clothing"] * 6 + ["Grocery"] * 6,
    "Units_Sold": [40, 35, 50, 45, 55, 60,
                   120, 110, 130, 140, 125, 150,
                   400, 420, 410, 430, 450, 460],
    "Revenue": [200000, 175000, 250000, 225000, 275000, 300000,
                96000, 88000, 104000, 112000, 100000, 120000,
                80000, 84000, 82000, 86000, 90000, 92000],
    "Profit": [30000, 24500, 40000, 31500, 44000, 51000,
               19200, 15840, 22880, 25760, 20000, 28800,
               8000, 9240, 7380, 9460, 9900, 11040],
}
df = pd.DataFrame(data)
df.to_csv("sales_data.csv", index=False)

print("===== DATA (first 6 rows) =====")
print(df.head(6))

# ---------- Step 2: Analysis ----------
# Q1. Category wise total revenue and profit
category_total = df.groupby("Category")[["Revenue", "Profit"]].sum()
print("\n===== Q1: Category wise Total =====")
print(category_total)

# Q2. Month wise total revenue
month_order = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
month_total = df.groupby("Month")["Revenue"].sum().reindex(month_order)
print("\n===== Q2: Month wise Revenue =====")
print(month_total)
print("Best Month:", month_total.idxmax())

# Q3. Profit margin % of each category
category_total["Profit_Margin_%"] = (category_total["Profit"] / category_total["Revenue"] * 100).round(2)
print("\n===== Q3: Profit Margin % =====")
print(category_total["Profit_Margin_%"])
print("Highest margin category:", category_total["Profit_Margin_%"].idxmax())

# ---------- Step 3: Charts ----------
# Chart 1: Category wise revenue
category_total["Revenue"].plot(kind="bar", color=["#e67e22", "#2980b9", "#27ae60"])
plt.title("Category wise Total Revenue")
plt.ylabel("Revenue (Rs)")
plt.xticks(rotation=0)
plt.tight_layout()
plt.savefig("chart1_category_revenue.png")
plt.close()

# Chart 2: Monthly revenue trend
month_total.plot(kind="line", marker="o", color="purple")
plt.title("Monthly Revenue Trend")
plt.ylabel("Revenue (Rs)")
plt.grid(True)
plt.tight_layout()
plt.savefig("chart2_monthly_trend.png")
plt.close()

# Chart 3: Profit margin
category_total["Profit_Margin_%"].plot(kind="bar", color="teal")
plt.title("Profit Margin % by Category")
plt.ylabel("Profit Margin (%)")
plt.xticks(rotation=0)
plt.tight_layout()
plt.savefig("chart3_profit_margin.png")
plt.close()

print("\nCharts saved!")
```

6. Output

Console Output:

```
===== DATA (first 6 rows) =====
  Month     Category  Units_Sold  Revenue  Profit
0   Jan  Electronics          40   200000   30000
1   Feb  Electronics          35   175000   24500
2   Mar  Electronics          50   250000   40000
3   Apr  Electronics          45   225000   31500
4   May  Electronics          55   275000   44000
5   Jun  Electronics          60   300000   51000

===== Q1: Category wise Total =====
             Revenue  Profit
Category
Clothing      620000  132480
Electronics  1425000  221000
Grocery       514000   55020

===== Q2: Month wise Revenue =====
Month
Jan    376000
Feb    347000
Mar    436000
Apr    423000
May    465000
Jun    512000
Name: Revenue, dtype: int64
Best Month: Jun

===== Q3: Profit Margin % =====
Category
Clothing       21.37
Electronics    15.51
Grocery        10.70
Name: Profit_Margin_%, dtype: float64
Highest margin category: Clothing

Charts saved!
```

Chart 1: Category wise Total Revenue

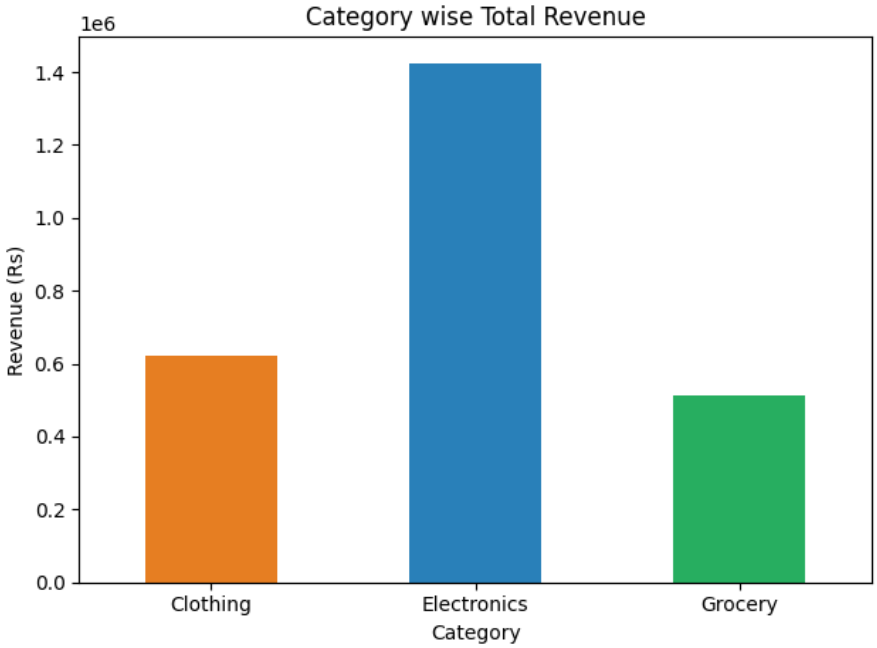


Chart 2: Monthly Revenue Trend

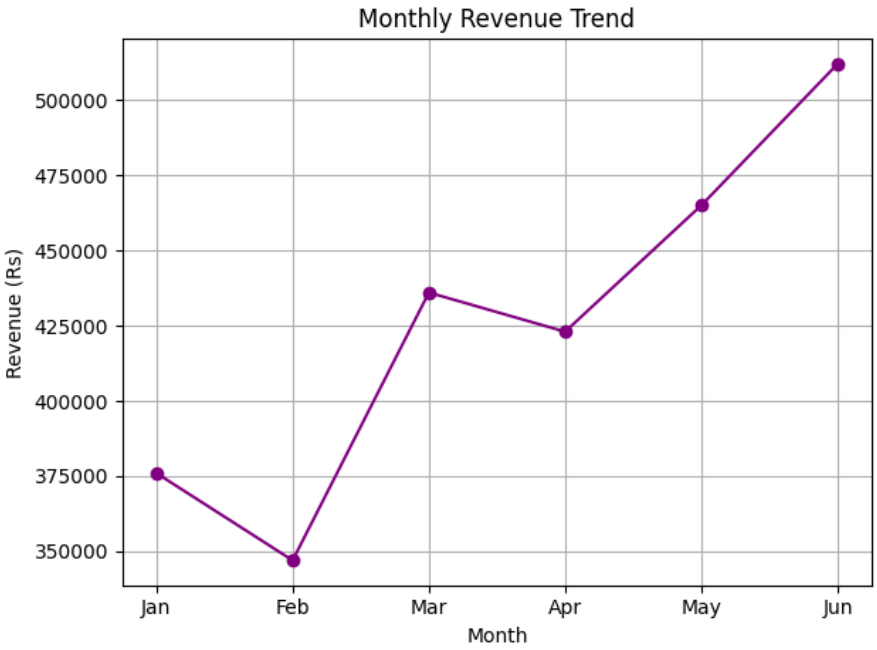
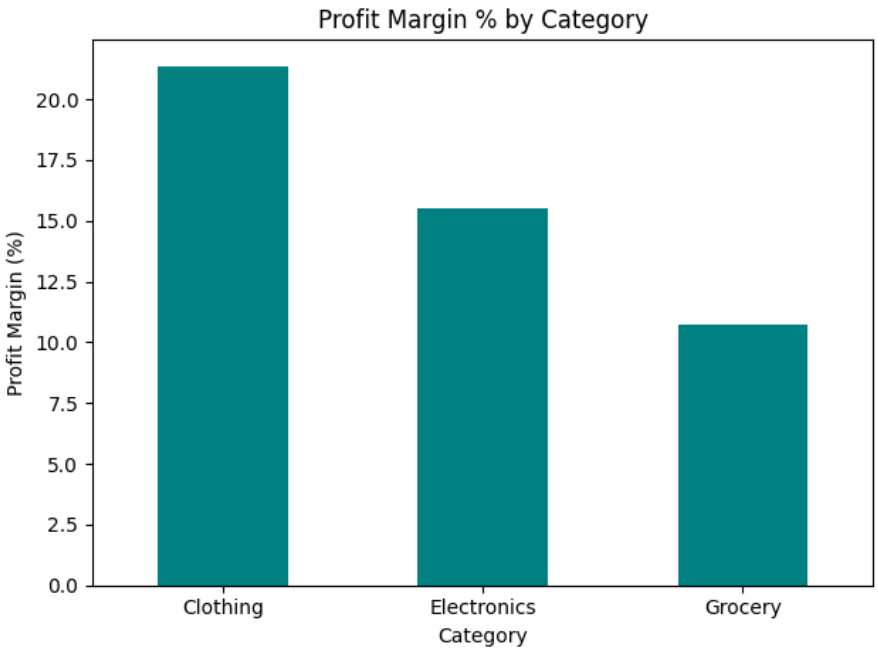


Chart 3: Profit Margin % by Category



7. Final Answer / Result

Question	Answer
Q1 Total Revenue	Electronics Rs 14,25,000 | Clothing Rs 6,20,000 | Grocery Rs 5,14,000
Q1 Total Profit	Electronics Rs 2,21,000 | Clothing Rs 1,32,480 | Grocery Rs 55,020
Q2 Best Month	June (Rs 5,12,000)
Q3 Highest Margin	Clothing (21.37%)

8. Conclusion

• Electronics sabse zyada Revenue aur Profit deta hai (total Revenue ka lagbhag 56%).
• Sales Feb me sabse kam thi aur Jun tak lagatar badhti gayi - overall trend upward hai.
• Clothing ka Profit Margin sabse zyada (21.37%) hai, isliye Clothing par promotion karna fayde ka ho sakta hai.
• Grocery me volume zyada hai lekin margin sabse kam (10.70%) hai.